Protocol – Harvesting and Precipitation of IPTG-Induced Bacteria

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

This protocol pellets the IPTG-induced cultures from Protocol 3, records the wet biomass of each pellet and stores the cells at −80 °C until lysis following Protocol 5.

Materials list

- Induced bacterial cultures
- Pre-weighed (tared) **50 mL Falcon tubes**
- Balance
- Distilled water (for balancing the centrifuge)
- Refrigerated centrifuge (**10 °C**)
- Filter paper or tissue (for draining)
- **−80 °C** freezer

Steps

1. Weigh the empty 50 mL Falcon tubes and record the tare.
2. Transfer the induced cultures from the flasks into the tared tubes and balance them in pairs with water for the centrifuge.
3. Centrifuge at **6000 ×g, 10 °C, for 10–15 min** to pellet the cells.
4. Discard the supernatant and drain the tubes by inversion to remove residual medium.
5. Weigh the tube with the pellet and calculate the biomass (pellet weight equals total weight minus the empty tube weight).
6. Freeze the cell pellets at **−80 °C** until further use (snap-freezing in liquid nitrogen is an alternative).

Troubleshooting

- **Biomass comparison:** Report wet biomass per culture volume (**g/L**) so that expression batches can be compared.
- **Loose pellet:** If the pellet is loose, remove the supernatant with a pipette instead of pouring it off.
- **Residual medium:** Drain the tubes fully before weighing, since residual medium inflates the biomass value.
- **Repeated freeze–thaw cycles:** Thaw each pellet only once, immediately before lysis.